DION LIU

AI Engineer

AI Engineer specializing in biomedical AI, building multi-omics and MRI-based systems from data pipeline to deployment.

me@dionic.xyz | github.com/xu62u4u6 | LinkedIn | Taiwan

EXPERIENCE

Hon Hai Precision Industry Co., Ltd. (Foxconn) — AI Engineer 2025.09 - Present

Designed and developed a multimodal deep learning classification system on MRI medical images for Alzheimer's disease progression prediction. Evaluated and adopted FastSurfer over FreeSurfer for brain segmentation, reducing per-subject preprocessing from hours to ~1 minute and enabling large-scale iterative experiments across 2,000+ subjects (10,000+ scans). Architected an experiment management platform (FastAPI + SSE + SQLite) supporting YAML-driven hyperparameter combinations, real-time training monitoring, Grad-CAM interpretability analysis, and NiiVue medical image viewing. Developed a digital care platform for neurodegenerative disease management — patient-facing cognitive tracking app (React), clinician dashboard (React), and FastAPI backend with SQLite. Integrated a RAG-based chatbot (LangChain) grounded in health education materials for caregiving guidance. Collaborated directly with clinicians to translate requirements into system specifications.

Gorigin Biomed Co., Ltd. — Bioinformatics Engineer 2021.10 - 2022.09

Built an NGS/qPCR gut microbiome analysis pipeline from scratch using Python, QIIME2, and pandas, deployed on GCP. Handled 16S amplicon preprocessing, taxonomy refinement via BLAST + NCBI, and automated report generation. First engineering role — transitioned from wet-lab biology to independent data pipeline development.

- Automated NGS/qPCR Reporting System (2021.10 - 2022.08)

EDUCATION

National Yang Ming Chiao Tung University — Master of Science in Biomedical Informatics 2023.09 - 2025.10

Graduate training centered on multi-omics analysis, data integration, and biomarker discovery using machine-learning approaches. Developed end-to-end analytical workflows—from pipeline construction to downstream processing, visualization, and result interpretation.

National Taiwan Ocean University — Bachelor of Science in Bioscience and Biotechnology 2017.09 - 2021.06

Studied organic chemistry, biochemistry, molecular biology, and standard wet-lab experimental techniques. Joined a structural bioinformatics laboratory, where I began working with protein structure analysis and gained my first experience in computational biology.

SKILLS

Bioinformatics: Bioinformatics, Genomics (DNA), Cancer Genomics, WES Pipelines, CNV Analysis, Mutational Sig., Transcriptomics, Bulk RNA, Spatial Tx, scRNA-seq, Methylation, Microbiomics, Multi-omics, Medical Imaging

Data Science: Data Analysis, Processing, Databases, EDA, Visualization

Machine Learning: Modeling, Model Usage, Deep Learning, Feature Eng., Validation, GenAI / LLM

Engineering & DevOps: Engineering, Linux / Ops, Version Control, CI/CD & Workflow, Packaging, Infrastructure

Web Development: Web Dev, Frontend, Backend

SELECTED PROJECTS

EnrichRAG github.com/xu62u4u6/EnrichRAG

A modular framework that augments gene set enrichment (GO/KEGG ORA) with knowledge graph retrieval (~39M relations from STRING, KEGG, Reactome, PubTator) and parallel literature search from PubMed and the web. An LLM pipeline extracts biomedical relations and synthesizes narrative reports, streamed in real time to a Vue 3 frontend with an interactive D3 network graph and result-grounded chat assistant.

pymaftools

github.com/xu62u4u6/pymaftools

Open-source Python cancer genomics toolkit published on PyPI (v0.4.1, 7 GitHub stars). Unified Cohort structure linking SNV, CNV, expression, and signature layers. Provides MAF parsing, statistical filtering with FDR correction, multi-omics stacking classifier with RFECV feature selection, and publication-ready visualizations (OncoPlot, LollipopPlot, PCA). Actively maintained 1.5+ years with external industry users.

ADNI MRI Alzheimer's Prediction

Multimodal deep learning system for Alzheimer's disease progression prediction from brain MRI. Implements image-only, tabular-only, and image+tabular fusion (ResNet50, CatBoost, late/middle fusion) with survival analysis. Includes a full preprocessing pipeline (DICOM → FastSurfer → MNI normalization), a FastAPI backend with job queue and SSE monitoring, and a browser-based dashboard with NiiVue viewer and Grad-CAM interpretability. Processed 2,000+ subjects and 10,000+ scans from ADNI.

dionic.xyz — A Bioinformatics Portfolio

A biomedical-themed portfolio built with Nuxt 3, where the visual narrative follows the story of cancer — from mutation to uncontrolled growth. Features a D3-powered skill graph inspired by Path of Exile, a git-graph experience timeline, terminal-aesthetic UI with DNA backgrounds, and an automated data layer that validates tag dependencies and generates a formatted CV PDF via Playwright in CI.

Genomic Landscape of Lung Adenosquamous Carcinoma

Comprehensive WES-based comparative genomics study of lung adenosquamous carcinoma (ASC) with lung adenocarcinoma and squamous cell carcinoma. Conducted SNV/INDEL, CNV, HRD, and mutational signature analyses, integrated multi-cohort datasets, and explored subtype- and component-level genomic patterns to investigate tumor evolution and cell-of-origin hypotheses.